Hover by Steering the Field Axis

Attitude by `vec=`, altitude by `freq=`: model, controller, identification and a stability certificate

ESP32_PMW `controller/control/axis_hover.py`, `axis_hover_proof.py`

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Abstract

We design a hover controller for the magnetically spun rotor. It steers the rotation axis of the coil field with the `vec=` command (`lib/VectorField`) to set the thrust direction, and it sets the field frequency with `freq=` to set the thrust magnitude. A PID on attitude is correct in spirit, but the attitude plant is *gyroscopic*: the rotor axis precesses about the field axis, with a phase of about 77°. A PID that tilts the field toward the wanted axis therefore drives the axis sideways. We write the plant in complex form, cancel the precession phase with an identified complex gain, and close the loop as a cascade PD with a disturbance observer. The observer predicts across the measured latency by rewinding and replaying the logged commands. We give a regression that identifies the commanded-to-actual axis map from flight video. We then prove robust Schur stability for 17–33 ms of latency over an explicit box of precession, gain and thrust errors, and separately over a line of residual axis leak. The proof uses a structured small-gain bound, exact H_∞ norms and branch-and-bound over the parameters. The joint four-parameter box is supported by exact eigenvalues at its corners and at 3000 random points, but it is not proven. We also bound the frequency excursion, so that the loop stays in a band far from the 190 Hz coil resonance, and we show with an energy argument that the commanded frequency slew cannot break phase lock. The main negative result is that the residual, position-dependent field-axis error (the "axis leak") sets the minimum lateral bandwidth. A slow, phase-robust design is destroyed by a leak of only 0.02°/mm.

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1 Verdict: is this the right way to hover?

Yes, with three changes to the plain-PID idea. All three are forced by the physics below.

1. **Attitude cannot be a PID on the tilt error.** The rotor axis obeys $\dot{\sigma} = \lambda(\nu - \sigma)$ with complex $\lambda = \Omega_p e^{\mathrm{j}\psi}$ and $\psi \approx -77°$ (§3). The tilt is a first-order, *rotating* response to the field tilt, not a double integrator. A D term on attitude adds nothing. Feeding back the tilt error with a real gain makes the axis spiral once $|\psi|$ approaches 90°. The fix is one complex multiply: command $\nu = \sigma + (k/\hat{\lambda})(\sigma_{\mathrm{ref}} - \sigma)$, so that the nominal closed attitude loop is $\dot{\sigma} = k(\sigma_{\mathrm{ref}} - \sigma)$ with real k.

2. **The integral terms are replaced by bias states in the observer.** A plain integrator overshot by 115 mm when the hover frequency was mis-estimated by 6% (§9). An estimated thrust bias and an estimated tilt bias give the same zero steady-state error for constant disturbances, and they act at observer speed.

3. **The lateral bandwidth has a floor, set by the axis leak.** When the robot is off-axis, the field axis tilts with position. Whatever part `vec=`'s geometry compensation does not remove acts as positive position feedback. The design tolerates $|\kappa| \le 0.004\,\mathrm{rad/mm}$ (0.23°/mm, about 16% of the modelled 1.4°/mm) and needs the precession estimate to within $\times[0.7, 1.4]$ and ±15°. Both numbers come from identification, so **identification is not optional**.

LQR or MPC on the same model (`design_hover_lqr.py`, `LLM/design/spatial_mpc.py`) would give similar gains. The cascade is preferred because each gain has a physical meaning, the law is three complex multiply-adds, and it can be certified by the method below. A frequency-only altitude loop, decoupled from attitude, is correct to first order. The coupling is second order in tilt and bounded in §6.5.

2 Notation

All lateral quantities are complex numbers. For a horizontal vector (a_x, a_y) we write $a = a_x + \mathrm{j}a_y \in \mathbb{C}$. A rotation by ϕ is multiplication by $e^{\mathrm{j}\phi}$. $\arg z = \mathrm{atan2}(\mathrm{Im}\, z, \mathrm{Re}\, z) \in (-\pi, \pi]$, where $\mathrm{atan2}(y, x)$ is the angle of the point (x, y) measured from the $+x$ axis. $\|\cdot\|$ is the spectral norm and $\|\cdot\|_F$ the Frobenius norm. $\rho(A)$ is the spectral radius. A^H is the conjugate transpose. $\det(M)$ is the determinant. Time steps are indexed $k \in \{0, 1, 2, \dots\}$.

symbol	meaning	unit
$w = x + \mathrm{j}y$	lateral offset from the hover point (datum frame)	mm
$v = \dot{w}$	lateral velocity	mm/s
$\sigma = (n_x + \mathrm{j}n_y)/n_z$	rotor-axis tilt, $\hat{\mathbf{n}}$ the measured unit axis with $n_z > 0$	rad
ν	field-axis tilt at the rotor (datum frame)	rad
ν_{cmd}	tilt sent in `vec=` (coil frame)	rad
z	height error, datum frame	mm
f, f_h	field frequency; frequency at which thrust equals weight	Hz
g	9810	mm/s^2
$\lambda = \Omega_p e^{\mathrm{j}\psi}$	precession gain of the rotor axis	rad/s
κ	axis leak: field tilt per mm of offset not compensated	rad/mm
$\beta = (f/f_h)^2$	thrust ratio	–
$T = 1/120$	control period	s
d, $\hat{d} = 3$	true and assumed measurement delay	steps

3 Plant

3.1 Rotor axis

The cycle-averaged torque of the rotating field on the spinning magnet has an aligning component perpendicular to the spin. Acting on the angular momentum $L = I\omega$, this component turns the spin axis *perpendicular* to both the spin axis and the field axis, which is precession. Slip and drag add a component toward the field axis. In tilt coordinates this gives

$$\dot{\sigma} = \lambda\,(\nu - \sigma), \qquad \lambda = \Omega_p e^{\mathrm{j}\psi}. \tag{1}$$

Pure precession is $|\psi| = 90°$, and pure alignment is $\psi = 0$. The multi-coil plant in `LLM/design/spatial_model.py`, with the axis re-allocated each step, has the eigenvalues $-0.78 \pm 3.43\mathrm{j}$ rad/s. The seed is therefore $\lambda_0 = 3.52\,e^{-\mathrm{j}77.2°}$ rad/s, with the sign of ψ set by the spin sense. The seed is a model value and must be measured (§5).

3.2 Field axis and the leak

`vec=x:y:z:q:b` solves for duty and phase so that the field at the commanded point rotates about $R(q)\hat{\mathbf{z}}$ (`theory.md` §28). The true axis differs from the commanded one through channel gain and phase trims, the coil-to-camera registration, and model error. We write

$$\nu = a\,\nu_{\mathrm{cmd}} + c + \kappa\,w, \tag{2}$$

where $a \in \mathbb{C}$ absorbs gain and yaw, $c \in \mathbb{C}$ is a constant offset (it includes the robot's crooked 1.6° seat), and κ is the residual dependence on position. Without compensation the field axis tilts by 1.4°/mm off-axis (`spatial_model`), and this is the open-loop divergence. The controller sends the *predicted robot position* as the `vec=` point, so VectorField removes most of the dependence, and κ is what remains.

3.3 Translation

Thrust is $mg\beta$ along the rotor axis. For small tilt,

$$\dot{w} = v, \qquad \dot{v} = g\beta\,\sigma, \qquad \ddot{z} = g\Big(\beta\cos|\sigma| - 1\Big), \qquad \beta = \big(f/f_h\big)^2. \tag{3}$$

The measured $f_h \approx 101\,\mathrm{Hz}$ (`theory.md` 18.17), and the f^2 law breaks down above about 110 Hz. We therefore keep $f \in [85, 115]\,\mathrm{Hz}$, which is $\geq 75\,\mathrm{Hz}$ below the coil resonance.

3.4 Model used by the controller

The lateral state is $x = (w, v, \sigma, b_\sigma) \in \mathbb{C}^4$, where b_σ is a constant tilt bias. The input is $u = \nu$, and the measurements are $y = (w, \sigma)$:

$$\dot{x} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & g & 0 \\ \hat{\lambda}\hat{\kappa} & 0 & -\hat{\lambda} & \hat{\lambda} \\ 0 & 0 & 0 & 0 \end{pmatrix} x + \begin{pmatrix} 0 \\ 0 \\ \hat{\lambda} \\ 0 \end{pmatrix} u. \tag{4}$$

The vertical state is $(z, \dot{z}, b_z) \in \mathbb{R}^3$ with input a_z (mm/s^2), $\ddot{z} = a_z + b_z$, and measurement z. Both models are discretised exactly (zero-order hold, block matrix exponential; `discretised`).

4 Controller

4.1 Lateral: complex cascade PD

Inner loop: $u = \sigma + (k/\hat{\lambda})(\sigma_{\mathrm{ref}} - \sigma) - b_\sigma - \hat{\kappa} w$, which gives $\dot{\sigma} = k(\sigma_{\mathrm{ref}} - \sigma)$ when the model is exact. Outer loop: $\sigma_{\mathrm{ref}} = -(K_p w + K_d v)/g$. The characteristic polynomial is

$$s^2(s + k) + k(K_d s + K_p) = s^3 + k s^2 + k K_d s + k K_p,$$

so the three poles p_1, p_2, p_3 fix $k = c_1$, $K_d = c_2/c_1$ and $K_p = c_3/c_1$, where $\prod_{i=1}^{3}(s - p_i) = s^3 + c_1 s^2 + c_2 s + c_3$ (`lateral_feedback`). The shipped poles are $(-12, -4 \pm 4\mathrm{j})\,\mathrm{rad/s}$. The outer natural frequency is 5.7 rad/s (0.9 Hz), and §9 explains why it cannot be lower. The state-feedback row is

$$K = \Big(\tfrac{c_3}{\hat{\lambda} g} + \hat{\kappa},\ \tfrac{c_2}{\hat{\lambda} g},\ \tfrac{c_1}{\hat{\lambda}} - 1,\ 1 \Big), \qquad u = -K\hat{x}.$$

The command is clamped to $|u| \leq 0.35\,\mathrm{rad}$. It is sent as $\nu_{\mathrm{cmd}} = (u - \hat{c})/\hat{a}$, as the axis $\hat{\mathbf{n}} \propto (\mathrm{Re}\,\nu_{\mathrm{cmd}}, \mathrm{Im}\,\nu_{\mathrm{cmd}}, 1)$ and the quaternion $q = \big(1 + n_z,\ -n_y,\ n_x,\ 0\big)/\|\cdot\|\cdot\|$. This is the minimal rotation $\hat{\mathbf{z}} \to \hat{\mathbf{n}}$, so the twist about the axis is identically zero and never jumps (`theory.md` §28.5).

4.2 Vertical: PD on acceleration, mapped to frequency

$a_z = -K_p z - K_d \dot{z} - b_z$, with poles $-2.22 \pm 2.22\mathrm{j}\,\mathrm{rad/s}$. The frequency that delivers a_z is

$$f = \hat{f}_h \sqrt{1 + a_z/g},$$

clamped to the band $[85, 115]\,\mathrm{Hz}$ and then slew-limited to 5 Hz/s. The *achieved* acceleration $g\big((f/\hat{f}_h)^2 - 1\big)$ is fed back into the observer, so saturation cannot wind up the bias.

4.3 Latency: rewind and replay

A pose stamped at host receipt t_{rx} was exposed at $t_{\mathrm{sh}} = t_{\mathrm{rx}} - 0.022\,\mathrm{s}$ (`constants.CAMERA_LATENCY_S`, measured from LED edges). At step k the controller holds its history $\{(\hat{x}_j, u_j)\}_{j=k-12}^{k-1}$. A measurement with age $\hat{d} = \mathrm{round}((t_k - t_{\mathrm{sh}})/T)$ is handled in two steps. First it corrects the stored estimate $\hat{x}_{k-\hat{d}} \leftarrow \hat{x}_{k-\hat{d}} + L(y - C\hat{x}_{k-\hat{d}})$. Then it replays $\hat{x}_{j+1} = \Phi\hat{x}_j + \Gamma u_j$ for $j = k - \hat{d}, \ldots, k - 1$,

overwriting the history (`DelayedLoop.correct`). This is a Kalman filter on the delayed time line plus an exact model predictor, and it is the feed-forward that interpolating between pose estimates cannot give: interpolation adds no information and adds a sample of lag, while the replay uses the commands that were actually applied. L is the steady-state Kalman gain from the discrete Riccati equation. Its noise levels are: position 0.034 mm (`noise_model.json`); tilt 0.1 rad, on purpose 25× the measured 0.0039 (§9); acceleration 30 mm/s^2; tilt rate 0.3 rad/s; bias drift 0.01 rad/s and 300 mm/s^3.

5 Identifying the commanded-to-actual axis map from video

Combining (1) and (2) gives a model that is linear in its unknowns:

$$\dot{\sigma} = \underbrace{\lambda a}_{\alpha}\,\nu_{\mathrm{cmd}} \;\underbrace{-\lambda}_{\beta_\sigma}\,\sigma \;+\; \underbrace{\lambda\kappa}_{\delta}\,w \;+\; \underbrace{\lambda c}_{\gamma}. \tag{5}$$

Given the measured axis $\sigma(t)$, offset $w(t)$ and the held command $\nu_{\mathrm{cmd}}(t)$, one complex least-squares solve returns $(\alpha, \beta_\sigma, \delta, \gamma)$, and then $\hat{\lambda} = -\beta_\sigma$, $\hat{a} = \alpha/\hat{\lambda}$, $\hat{\kappa} = \delta/\hat{\lambda}$ and $\hat{c} = \gamma/\hat{\lambda}$ (`fit_axis_response`). $\dot{\sigma}$ and σ come from a quadratic Savitzky–Golay filter (15 samples) on a uniform resample. The coil poses enter only through $\hat{a}$, $\hat{c}$ and the `vec=` point offset. If the registration is wrong, $\hat{a}$ and $\hat{c}$ absorb the error.

Experiment. `fly(law="axis", identify_axis=True, ...)` holds altitude closed-loop and sends open-loop tilt steps of 0.05 rad: zero, then eight azimuths, each for 1.5 s, with zero in between. It writes `<run>_axis.npz`. Then run `python controller/control/axis_hover.py fit <run>_axis.npz response.json` and fly with `axis_response="response.json"`. The runner refuses `law="axis"` without a fitted response.

Cautions. The video axis is unsigned, so the fit pins $n_z > 0$. Spin rate and spin sense are not observable above about 26 Hz (blade strobe), so the sign of ψ is a fit output, not an input. Takes with dropped frames are mistimed and must be excluded. Stay clear of the singular shell at $z \approx 20.8 \pm 1.5$ mm above the coils, where $a \to 0$; the controller refuses hover points there. If the rod constrains the tilt on the stand, run the steps in free hover with the seed and a small amplitude instead. The same regression works on closed-loop flight logs, since ν_{cmd} is logged.

6 Stability certificate

6.1 The closed loop as an exact linear recursion

Let the true lateral plant ((1)–(3), bias removed, since a constant disturbance does not affect stability) have the ZOH step $[\Phi_p\ \Gamma_p](\theta)$ for the parameters $\theta = (|\lambda/\lambda_0|,\ \arg(\lambda/\lambda_0),\ \beta,\ \kappa)$. The augmented state is

$$\zeta_k = \big(x^p_k,\, x^p_{k-1}, \ldots, x^p_{k-d};\ e_k;\ u_{k-\hat{d}}, \ldots, u_{k-1}\big),$$

where e_k is the stored estimate in the slot that the incoming measurement corrects. One step of `DelayedLoop` is then exactly

$$\zeta_{k+1} = \mathcal{A}(\theta)\zeta_k, \qquad \mathcal{A}(\theta) = \mathcal{A}_0 + E\,[\Phi_p\ \Gamma_p](\theta)\,F,$$

with E inserting the new plant state and F extracting (x^p_k, u_k) (`closed_loop`). The map is affine in the plant step. That structure is what makes the proof tractable.

6.2 Theorem

Theorem 1. *Let the true delay be $d \in \{2,3,4\}$ steps (16.7–33.3 ms), with $\hat{d} = 3$ assumed. Then $\rho(\mathcal{A}(\theta)) < 1$ for every constant θ in each of the following sets:*

(a) $\Theta_1 = [0.7, 1.4] \times [-15°, 15°] \times [0.77, 1.26] \times \{0\}$*, which covers precession gain, precession phase and thrust ratio with the leak identified and cancelled;*

(b) $\Theta_2 = \{1\} \times \{0\} \times \{1\} \times [-0.004, 0.004]$ *rad/mm, which covers the residual leak at the nominal precession.*

The vertical loop is Schur stable for every $\beta \in [0.5, 2]$ and $d \in \{2,3,4\}$, so f_h may lie anywhere in $[71, 143]$ Hz around the assumed 101 Hz.

The joint box $\Theta_1 \times [-0.004, 0.004]$ is *not* proven. Covering it would take about 7×10^5 cells, roughly 30 h, because the small-gain bound treats the real parameter κ as if it were complex. The worst spectral radius over its 2^4 corners and 3000 uniform samples, at all three delays, is 0.99938.

Proof. Step 1 (one stable point). At the centre θ_c of a cell, $\rho(\mathcal{A}(\theta_c)) < 1$ is checked directly.

Step 2 (no eigenvalue crosses the unit circle). By continuity of the eigenvalues, stability holds on a connected cell if $\det(zI - \mathcal{A}(\theta)) \neq 0$ for all $|z| = 1$ and all θ in the cell. Write $\mathcal{A}(\theta) = \mathcal{A}_c + E\Delta F$ with $\Delta = [\Phi_p \Gamma_p](\theta) - [\Phi_p \Gamma_p](\theta_c)$ and $G(z) = F(zI - \mathcal{A}_c)^{-1}E$. Since $zI - \mathcal{A} = (zI - \mathcal{A}_c)(I - (zI - \mathcal{A}_c)^{-1}E\Delta F)$, it suffices that $\sup_{|z|=1} \|G(z)\Delta\| < 1$, which is the small-gain condition.

Step 3 (structured bound on Δ). Let $X(\theta) = T\,DM(\theta)D^{-1}$ be the scaled generator of the ZOH block, with D the diagonal scaling $(|\lambda_0|^2/g,\ |\lambda_0|/g,\ 1,\ 1)$. The scaling is a similarity, so it leaves the stability question unchanged and balances mm against rad. With $\mathrm{L}_X(\cdot)$ the Fréchet derivative of exp at X_c, and $Y = X(\theta) - X_c = \sum_i (\theta_i - \theta_{c,i})\partial_i X + Y_2$,

$$\Delta = \sum_i (\theta_i - \theta_{c,i})\,\mathrm{L}_X(\partial_i X) + \mathrm{L}_X(Y_2) + R, \qquad \|R\| \le \tfrac{1}{2}\|Y\|^2 e^{\|X_c\| + \|Y\|}.$$

$\mathrm{L}_X(\partial_i X)$ is the upper-right block of $\exp\left(\begin{smallmatrix} X_c & \partial_i X \\ 0 & X_c \end{smallmatrix}\right)$. Y_2 collects the curvature of $\lambda = \lambda_0 r e^{\mathrm{j}\phi}$ and of the product $\lambda\kappa$, bounded entrywise by $|\lambda_0|(h_r h_\phi + \frac{1}{2} r_{\max} h_\phi^2)$ and its product rule (`LateralUncertainty.curvature`). Hence

$$\sup_{|z|=1} \|G\Delta\| \le \sum_i h_i \,\|G\,\mathrm{L}_X(\partial_i X)\|_\infty + \|G\|_\infty \Big(\|Y_2\| e^{\|X_c\|} + \tfrac{1}{2}\|Y\|^2 e^{\|X_c\| + \|Y\|} \Big),$$

where h_i are the cell half-widths. The certificate requires the right-hand side to be < 1.

Step 4 (exact H_∞ norms). Each $\|\cdot\|_\infty$ is computed with the Cayley transform $z = (1+s)/(1-s)$, which maps the unit circle onto the imaginary axis and preserves the peak gain. It is followed by the Hamiltonian test (Boyd–Balakrishnan): $\gamma > \|G\|_\infty$ if and only if the Hamiltonian $H(\gamma)$ has no imaginary eigenvalue. Bisection keeps an upper end γ at which no crossing is found (`peak_gain`). A dense unit-circle grid agrees with it to 0.5% and sits on the safe side.

Step 5 (cover). Each set is split along its widest normalised side until every cell satisfies Steps 1–3 (`certify`). For $d = 2, 3, 4$ the run needed: Θ_1: 464, 476, 610 cells; Θ_2: 13, 13, 16 cells; vertical: 1, 1, 1 cell(s). □

What the theorem does not cover. (i) The arithmetic is IEEE double, not interval arithmetic, so this is a computer-assisted proof up to rounding. The reported slacks exceed rounding by many orders of magnitude, but they are not formally enclosed. (ii) Parameters are frozen. Slow variation is covered by the usual slowly-varying argument; fast switching, such as frame dropouts (a missing correction), is not. (iii) The plant is the linearisation; saturation is handled in §6.3 and phase lock in §6.4. (iv) The leak is certified on its own, not jointly with the other three errors, and it is modelled as isotropic, a real κ along w. An anisotropic or rotated leak (a general real 2×2 map) needs another dimension. A real-parameter (μ) bound would remove the complex-κ conservatism and is the natural next step. (v) The pose filter that feeds `tick.pose` is not in the model; at $500\,\mathrm{mm/s^2}$ process noise it is nearly transparent at 1 Hz.

6.3 Staying in the band: no saturation for small errors

For the nominal loop we iterate $\zeta_{k+1} = \mathcal{A}\zeta_k$ from a consistent initial offset (plant and estimate agree) until it decays below 10^{-6}. We record the peak field-tilt command and the peak $|a_z|$ and $|\Delta a_z|$ per mm of initial offset. By linearity:

- a lateral offset of up to 31.47 mm never reaches the 0.35 rad tilt clamp;
- a height error of up to 290.35 mm never leaves $[85, 115]$ Hz, and up to 21.99 mm never meets the 5 Hz/s slew limit;
- the frequency moves by at most 0.05 Hz per mm of height error.

The loop therefore never commands within 75 Hz of the coil resonance. In simulation the steady frequency ripple is 0.34 Hz peak-to-peak.

6.4 Phase lock under the frequency slew

Lemma 1. *Let the lag δ obey $I\ddot{\delta} = \tau(\ell - \sin\delta)$ with load $\ell = (k_{\mathrm{drag}} f^2 + 2\pi I \dot{f})/\tau$, and let $\dot{f}$ switch from $-\dot{f}_{\max}$ to $+\dot{f}_{\max}$ (the worst slew reversal). Lock is kept if $U(\pi - \arcsin\ell_+) > U(\arcsin\ell_-)$, with $U(\delta) = -\cos\delta - \ell_+\delta$.*

Proof. U is the potential of the post-switch dynamics. The rotor starts at rest relative to the old equilibrium $\arcsin\ell_-$ with energy $U(\arcsin\ell_-)$. The separatrix passes through the saddle $\pi - \arcsin\ell_+$. With no damping (a conservative assumption) the orbit is bounded inside the lock basin if and only if its energy is below the saddle. □

With `z_track.TorqueLimits` at $f_h = 101$ Hz, $\dot{f}_{\max} = 5$ Hz/s and every f in the band, the margin is 1.239 (> 0). It stays positive down to $0.268\times$ the modelled peak torque. The slew term is negligible: $2\pi I \dot{f}_{\max} \approx 0.5\%$ of the peak torque. What matters is the drag at the top of the band. The modelled torque depends on the stale $F_{\mathrm{STEPOUT}} = 225$ Hz, so this lemma says how wrong the step-out torque may be (up to a factor $1/0.268$) before the band must shrink.

6.5 Coupling between attitude and altitude

Tilt reduces vertical thrust by $g\beta(1 - \cos|\sigma|) \le g\beta|\sigma|^2/2$. At the 0.35 rad clamp this is $\le 6\%\, g$; at the ≤ 0.02 rad of steady hover it is $0.02\%\, g$. The vertical loop is exponentially stable and its bias state absorbs the constant part. The remainder is quadratic in a state that decays exponentially, so the cascade is locally exponentially stable. Frequency changes β in the lateral loop, and Theorem 1 covers $\beta \in [0.77, 1.26]$, which is $f \in [89, 113]$ Hz at $f_h = 101$.

7 Latency budget and the pose pipeline

stage	time	source
exposure → host receipt (USB, MJPEG decode, AVFoundation)	16–28 ms	LED edges, `LLM/analyse/start.`
segmentation + estimation (native tracker)	3.3 ms	`pose/theory.md` 22.5
control step + `vec=` on the wire at 921600 baud	1.3 ms	`theory.md` 19.13
firmware solve + apply	< 1 ms	–
total	21–33 ms	covered: $d \in \{2, 3, 4\}$

Is the 120 Hz requirement met? The native tracker (C++ capture, interleaved pairing) delivers 221–240 poses/s on the live rig with zero lost out of 34 700. The hover runner, however, was calling `stereo_frames` without `tracker=True`, and so flew on the Python path at 65–100 Hz. **Changed:** `RunConfig.tracker=True` by default. The controller runs at a fixed 120 Hz on the newest pose, which leaves 1.8× headroom.

The largest latency term was not compensated. `tick.t` is the host receipt time, not the shutter time, so the existing predictor propagated only about 3 ms. **Changed:** both the LQR path and the axis law now use $t_{\text{sh}} = t_{\text{rx}} - 22$ ms. The constant was moved to `control/constants.py` and is shared with the video alignment.

Compute. Changed: `live_viz` no longer calls `trk.stats()` on every pose. That call copies up to 20 000 skew samples and sorts them under the GIL; it now runs once a second. **Left as measured opportunities** (each needs a native rebuild and a parity run):

1. `segment()` runs absdiff, thresholds, morphology and connected components on the full frame even when the ROI is known. It is 82% of the 3.3 ms. An ROI window would roughly halve the cost per pose.
2. a full-frame `float32` buffer is allocated per call in `ring_weight`, the structuring element is rebuilt per call, and there are five full-frame passes in `RunningPlate::update`;
3. a `std::thread` is created per pose when both views are fresh (`stereo.cpp:240`);
4. both frames are deep-cloned per worker step, and `result_py` copies about 1 MB of masks into Python per pose;
5. `best_format` may select MJPEG. A raw (YUYV/GREY) mode would remove the host decode, about 6 ms of CPU, and part of the 22 ms latency, if the USB bandwidth allows it at 640×400.

None of these is needed for hover at 120 Hz. The latency in item 5 is the only one that would move the certificate: it lowers d.

Why not a faster control clock? The loop poles are at most 12 rad/s. At $T = 1/120$ s, ZOH costs $\omega T/2$ of phase: 1.4° at the 5.7 rad/s outer crossover and 2.9° at the inner pole. A 500 Hz clock would recover about 2° of that, consistent with `theory.md` 19.14. The outer loop sits at 0.9 Hz, just above the old 0.8 Hz rule. That rule came from one uncompensated frame of delay on the az/mag plant; here the delay is predicted across and certified.

8 Disturbance rejection, in simulation

`axis_hover.simulate` implements (1)–(3) with ground contact, measurement noise (0.034 mm, 0.0039 rad) and a pure delay. Results over 40 s:

case	lateral RMS	height RMS	overshoot z	f range
nominal, mid-air start, 25 ms	0.13 mm	0.13 mm	0.65 mm	100.8–101.2 Hz
takeoff from rest, true $f_h = 94$ Hz (-7%)	0.13 mm	0.13 mm	0.62 mm	91.2–94.2 Hz
takeoff from rest, true $f_h = 108$ Hz ($+7\%$)	0.13 mm	0.13 mm	0.68 mm	103.7–108.2 Hz
crooked seat, $c = 0.028$ rad (1.6°)	0.13 mm	0.13 mm	0.65 mm	98.0–101.2 Hz
corner: $\lambda \times 1.3e^{\mathrm{j}15^\circ}$, $\kappa = 0.004$, 33 ms	0.19 mm	0.14 mm	0.66 mm	100.8–101.2 Hz

The self-check also flies six random plants (gain 0.75–1.35, phase $\pm 18^\circ$, $|\kappa| \le 0.005$, $f_h \in [92, 110]$ Hz, 16–33 ms, takeoff from rest), and it recovers a synthetic (λ, a, c, κ) from a tilt-step take to within 10%.

9 Negative results

1. **Integral action overshoots.** PID with integrators, started mid-air with f_h off by 6%, climbed 115 mm before the integrator caught up. Replaced by bias states.

2. **Trusting the tilt measurement costs a millimetre.** A tilt error e produces a lateral acceleration ge, and $g/\omega^2 \approx 3400$ mm/rad at 0.27 Hz. With the measured 0.0039 rad per frame, the hover wandered by about 1 mm RMS whatever the process noise. Weighting the tilt measurement at 0.1 rad lets position, which is g-exact, carry the tilt estimate.

3. **The slow, phase-robust design fails on the leak.** Poles $(-6, -1.2 \pm 1.2\mathrm{j})$ tolerate $\pm 40^\circ$ of precession phase and a $\times[0.5, 2]$ gain error. They tolerate only $|\kappa| \le 3.6 \times 10^{-4}$ rad/mm (0.02°/mm), because the outer stiffness $K_p/g = 2 \times 10^{-4}$ rad/mm is 35× smaller than a leak of 30% of the modelled 1.4°/mm. Trade-off (phase tolerance at gains $[0.7, 1.4]$, leak tolerance):

poles (rad/s)	phase	$\lvert\kappa\rvert_{\max}$ (rad/mm)
$-9,\ -3 \pm 3\mathrm{j}$	30°	0.0031
$-12,\ -4 \pm 4\mathrm{j}$ (shipped)	20°	0.0059
$-15,\ -5 \pm 5\mathrm{j}$	10°	0.0112

   The certified box uses 15° and 0.004 to leave room at the corners.

4. **A common quadratic Lyapunov function and an unstructured small gain are too conservative.** With P from $\mathcal{A}^H P \mathcal{A} - P = -I$, the invariant ellipsoid admitted only 0.006 mm. The unstructured complex stability radius (8×10^{-6} before scaling, 10^{-3} after) needed millions of cells. The per-direction H_∞ bound of Step 3 is about 100× tighter.

10 Unmeasured, in order of what blocks hover

1. λ (rate and phase, sign by spin sense), a, c and κ: run the identification take.

2. The coil-to-camera registration. The holder survey is unconfirmed; this report assumes the coil pose is known (`coil_offset_mm`, `coil_yaw_deg`).

3. f_h to $\pm 7\%$. It is certified over a much wider range, but the band $[85, 115]$ assumes it.

4. The step-out torque. The lemma tolerates down to $0.268\times$ the model value.

5. The singular-shell height. The controller refuses 19.3–22.3 mm; the true height must be measured.

6. The VectorField `scale=` reply is not yet read back. If it drops below 1, the field is weaker than commanded. Land on it once a flight shows it happening.